Lab 3: Relational Algebra

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1 Queries in Relational Algebra

1. *Find the first and last names of persons who have made at least one equipment reservation.*

   $\pi_{\text{firstName, lastName}}\,(\text{Person} \bowtie_{\text{pid=madeByPid}} \text{Reservation})$

2. *Find the IDs of persons who are either attached to a laboratory or act as PI of at least one research project.*

   $\pi_{\text{pid}}\,(\text{Attached}) \cup \pi_{\text{pid}}\,(\sigma_{\text{role='PI'}}\,(\text{Participates}))$

3. *Find the IDs of laboratories located in the building 'College of Computing' or containing at least one equipment unit whose model belongs to the category 'CPU Server'.*

$$\begin{aligned}&\pi_{\text{labId}}\big(\sigma_{\text{building='College of Computing'}}(\text{Laboratory})\big)\\ \cup\ &\pi_{\text{labId}}\Big(\text{EquipmentUnit} \bowtie_{\text{modelId=modelId}} \big(\sigma_{\text{category='CPU Server'}}(\text{EquipmentModel})\big)\Big)\end{aligned}$$

4. *Find the IDs of laboratories that contain at least one 'CPU Server' unit and at least one 'GPU Server' unit.*

$$\begin{aligned}&\pi_{\text{labId}}\Big(\text{EquipmentUnit} \bowtie_{\text{modelId=modelId}} \big(\sigma_{\text{category='CPU Server'}}(\text{EquipmentModel})\big)\Big)\\ \cap\ &\pi_{\text{labId}}\Big(\text{EquipmentUnit} \bowtie_{\text{modelId=modelId}} \big(\sigma_{\text{category='GPU Server'}}(\text{EquipmentModel})\big)\Big)\end{aligned}$$

5. *Find the Faculty members who supervise every laboratory located in the building 'College of Computing'.*

   $\pi_{\text{supervisorPid, labId}}\,(\text{Laboratory}) \,/\, \pi_{\text{labId}}\,(\sigma_{\text{building='College of Computing'}}\,(\text{Laboratory}))$

6. *Find the IDs of persons who participate in every research project funded by at least one budget that also funds the laboratory with labId = 2.*

$$\begin{aligned}R_1 &= \pi_{\text{budgetLine}}\big(\sigma_{\text{labId=2}}(\text{FundsLab})\big)\\ R_2 &= \pi_{\text{projectCode}}\big(\text{FundsPrj} \bowtie_{\text{budgetLine=budgetLine}} R_1\big)\\ &\quad \pi_{\text{pid, projectCode}}(\text{Participates}) \,/\, R_2\end{aligned}$$

7. *Find pairs of persons (p_1, p_2) such that p_1 has made more reservations than p_2.*

$$\begin{aligned}R_1 &= {}_{\text{madeByPid}}\gamma_{\text{COUNT(resId)}\to n_1}(\text{Reservation})\\ R_2 &= {}_{\text{madeByPid}}\gamma_{\text{COUNT(resId)}\to n_2}(\text{Reservation})\\ R_1 &= \rho_{\text{id1}\leftarrow\text{madeByPid}}(R_1)\\ R_2 &= \rho_{\text{id2}\leftarrow\text{madeByPid}}(R_2)\\ &\quad \pi_{\text{id1, id2}}\big(\sigma_{n_1>n_2}(R_1 \times R_2)\big)\end{aligned}$$

8. *Find the IDs of persons who have reserved equipment units located in at least two different laboratories.*

$$
\begin{aligned}
R_1 &= \pi_{\text{madeByPid, labId}}\Big(\text{Reservation} \bowtie_{\text{resId=resId}} \big(\text{Reserves} \bowtie_{\text{serialNo=serialNo}} \text{EquipmentUnit}\big)\Big)\\
R_2 &= \rho_{\text{madeByPid2, labId2}}(R_1)\\
&\quad \pi_{\text{madeByPid}}\Big(\sigma_{(\text{madeByPid=madeByPid2})\wedge(\text{labId}\neq\text{labId2})}(R_1 \times R_2)\Big)
\end{aligned}
$$

9. *Find the IDs of reservations submitted in September 2026 that reserve at least one equipment unit currently located in the laboratory named 'AI Movement'.*

$$
\begin{aligned}
&\pi_{\text{resId}}\Big(\sigma_{\text{submissionTS}\geq\text{'2026-09-01 00:00:00'}\,\wedge\,\text{submissionTS}<\text{'2026-10-01 00:00:00'}}(\text{Reservation})\\
&\qquad \bowtie_{\text{resId=resId}} \Big(\text{Reserves} \bowtie_{\text{serialNo=serialNo}}\\
&\qquad \big(\text{EquipmentUnit} \bowtie_{\text{labId=labId}} (\sigma_{\text{name='AI Movement'}}(\text{Laboratory}))\big)\Big)\Big)
\end{aligned}
$$

10. *Find the IDs of persons who act as PI of more than one research project.*

$$
\begin{aligned}
R_1 &= \pi_{\text{pid, projectCode}}\big(\sigma_{\text{role='PI'}}(\text{Participates})\big)\\
R_2 &= \rho_{\text{pid2, projectCode2}}(R_1)\\
&\quad \pi_{\text{pid}}\Big(\sigma_{(\text{pid=pid2})\wedge(\text{projectCode}\neq\text{projectCode2})}(R_1 \times R_2)\Big)
\end{aligned}
$$

11. *Find the IDs of approvers who have approved reservations involving equipment units located in more than one laboratory.*

$$
\begin{aligned}
R_1 &= \pi_{\text{approverPid, labId}}\Big(\sigma_{\text{approverPid}\neq\text{NULL}}(\text{Reservation})\\
&\qquad \bowtie_{\text{resId=resId}} \text{Reserves} \bowtie_{\text{serialNo=serialNo}} \text{EquipmentUnit}\Big)\\
R_2 &= \rho_{\text{approver2, lab2}}(R_1)\\
&\quad \pi_{\text{approverPid}}\Big(\sigma_{(\text{approverPid=approver2})\wedge(\text{labId}\neq\text{lab2})}(R_1 \times R_2)\Big)
\end{aligned}
$$

12. *Find the IDs of persons who have no reservation whose planned reservation interval includes November 6, 2026.*

$$
\begin{aligned}
R_1 &= \pi_{\text{madeByPid}}\Big(\sigma_{\text{plannedStart}<\text{'2026-11-07 00:00:00'}\,\wedge\,\text{plannedEnd}\geq\text{'2026-11-06 00:00:00'}}(\text{Reservation})\Big)\\
R_2 &= \rho_{\text{pid}}(R_1)\\
&\quad \pi_{\text{pid}}(\text{Person}) - R_2
\end{aligned}
$$

13. *Find the laboratories whose number of equipment units is below the average number of equipment units per laboratory.*

$$
\begin{aligned}
R_{\text{counts}} &= {}_{\text{labId}}\gamma_{\text{COUNT(serialNo)}\rightarrow\text{NbEqui}}(\text{EquipmentUnit})\\
R_{\text{avg}} &= \gamma_{\text{AVG(NbEqui)}\rightarrow\text{avg}}(R_{\text{counts}})\\
&\quad \pi_{\text{labId}}\big(\sigma_{\text{NbEqui}<\text{avg}}(R_{\text{counts}} \times R_{\text{avg}})\big)
\end{aligned}
$$

14. *For each laboratory, return the person who has made the greatest number of distinct approved reservations involving equipment units located in that laboratory.*

$$R_{\text{counts}} = {}_{\text{labId, madeByPid}}\gamma_{\text{COUNT(DISTINCT resId)}\to\text{totalRes}}\Big(\sigma_{\text{approverPid}\neq\text{NULL}}(\text{Reservation}) \bowtie_{\text{resId=resId}} \text{Reserves} \bowtie_{\text{serialNo=serialNo}} \text{EquipmentUnit}\Big)$$

$$R_{\max} = {}_{\text{labId}}\gamma_{\text{MAX(totalRes)}\to\text{maxRes}}(R_{\text{counts}})$$

$$\pi_{\text{labId, madeByPid}}\big(\sigma_{\text{totalRes=maxRes}}(R_{\text{counts}} \bowtie_{\text{labId=labId}} R_{\max})\big)$$

15. *List the IDs of persons who made at least three equipment reservations during the year 2025.*

$$R_{\text{count}} = {}_{\text{madeByPid}}\gamma_{\text{COUNT(DISTINCT resId)}\to\text{totalRes}}\Big(\sigma_{\text{submissionTS}\geq\text{'2025-01-01 00:00:00'}\,\wedge\,\text{submissionTS}<\text{'2026-01-01 00:00:00'}}(\text{Reservation} \bowtie_{\text{resId=resId}} \text{Reserves})\Big)$$

$$\pi_{\text{madeByPid}}\big(\sigma_{\text{totalRes}\geq 3}(R_{\text{count}})\big)$$

2 Key Challenges and How We Addressed Them

1. every In the questions 5 and 6 what we did is that we first built the set of required elements (the College of Computing labs, or the projects funded by a budget line of lab 2) and divided the (person, element) relation by it.

2. Interval overlap. In question 12 what we decided to think a bit outside the box and that's because of one example let's suppose we directly checked for a person that has a reservation that starts after november 6 or that ends before november 6, that would mean that he is a free person, one that we out his id, but what if now he has a second reservation, and this time it overlapse the specific date, since he had one valid reservation beforehand that would mean that his id will be returned even though he has a reservation that overlaps, now what we did is that we identify the busy people and remove them fro all the people that way we will be kept with only the free people.

3. "At least two" / "more than one". Here we did something clever and a bit tricky,, we first create a table with the what is required and create a copy of it and then we do the multiplication between them (x) and for example retrieve the ids from that multiplication where the ids are the same but the laboratory differs.

4. Counting the same reservation more than once. In question 15 we noticed that when we join Reservation with Reserves, a reservation that books several equipment units appears once for every unit. For example, a person with only 2 reservations, one of them with 2 units, would end up with 3 rows, and a simple COUNT would wrongly say 3 and return them. To fix this we used COUNT(DISTINCT resId), so each reservation is counted only once no matter how many units it reserves.